Spectra Paints Inc.

Marketing Campaign Analysis using MySQL

Consumer growth · Campaign impact · Market basket analysis



Business context

Spectra Paints launched in 2019 and has grown steadily across tier 1 and 2 cities, now expanding into tier 3. Orders come from household, industrial and government sectors. As a data consultant, the goal is to help marketing with consumer growth, lead conversion, new campaigns and market basket analysis.

2019

Company launched

Fast-growing, tier 1-3 cities

3

Sectors served

Household, industrial, government

6

Tables analysed

Customers, items, transactions, coupons, cities, campaigns

Data and SQL toolkit

- sqlproject_customer
- sqlproject_item
- sqlproject_customertransactiondata
- sqlproject_couponmapping
- sqlproject_citydata
- sqlproject_campaign

Concepts applied

- Filtering with SQL operators
- Aggregation and sorting
- String, math and date-time functions
- JOINS and subqueries
- Window functions
- CASE, procedures, functions, views

Business snapshot

5

Paint categories

Oil, Anti-Corrosive, Enamel, Emulsion, Synthetic

2

Coupon types

Flat and Percent

21

States served

Where products are delivered

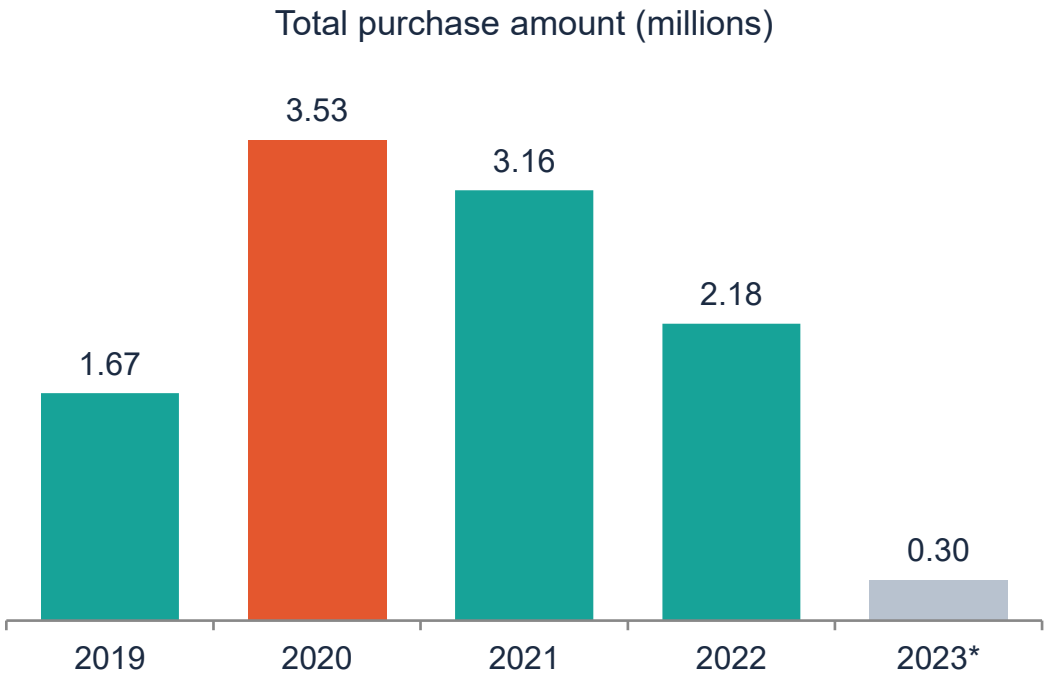
3

Order types

Household, Government, Industrial

```
SELECT COUNT(DISTINCT Item_Category) FROM sqlproject_item;    -- 5
```

Sales peaked in 2020, then declined



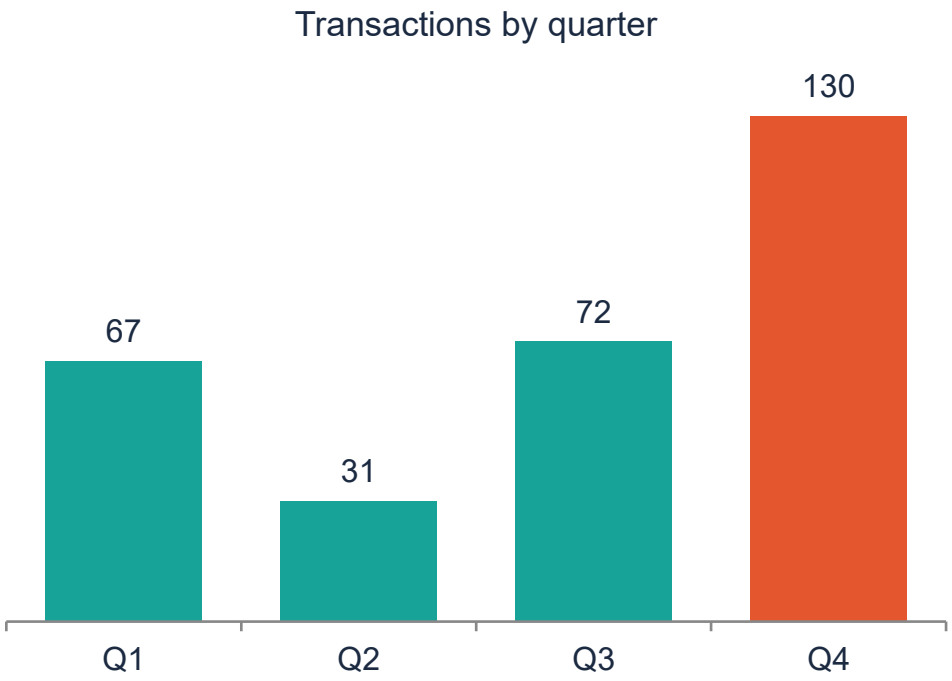
+111% 2020 vs 2019

-11% 2021 vs 2020

-31% 2022 vs 2021

Quantity followed the same pattern: 4,546 > 9,888 > 9,184 > 6,532 units. *2023 has March data only.

Q4 is the busiest season



130

Q4 transactions

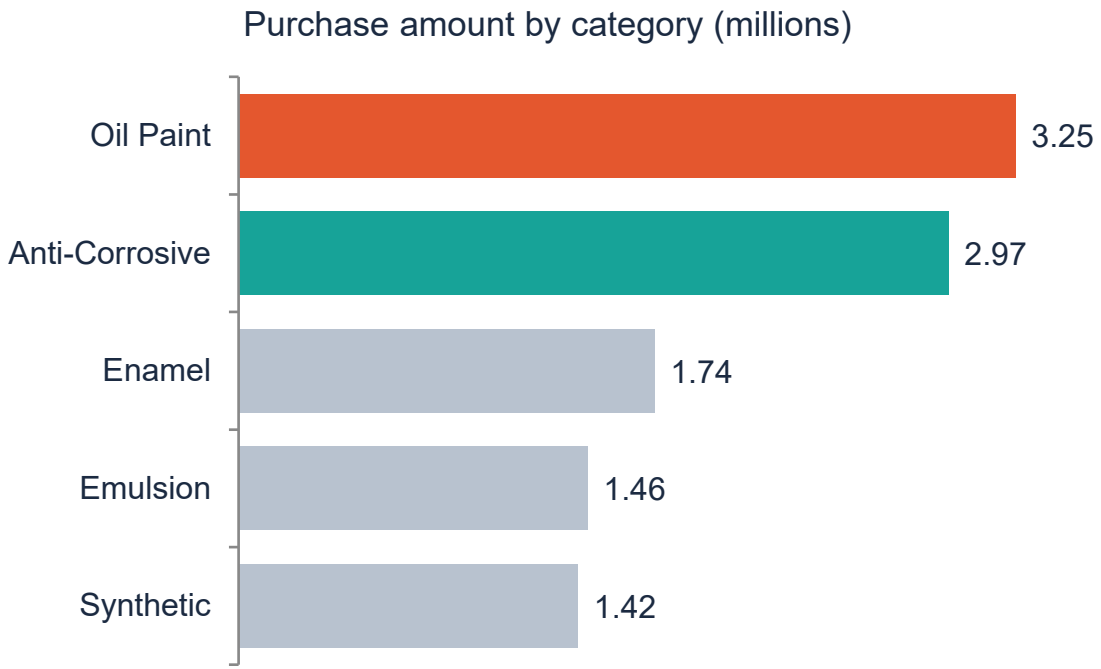
43% of all 300

48

Dec 2020 transactions

Busiest month overall

Oil Paint leads category sales



57%

Top 2 categories

Oil and Anti-Corrosive paints

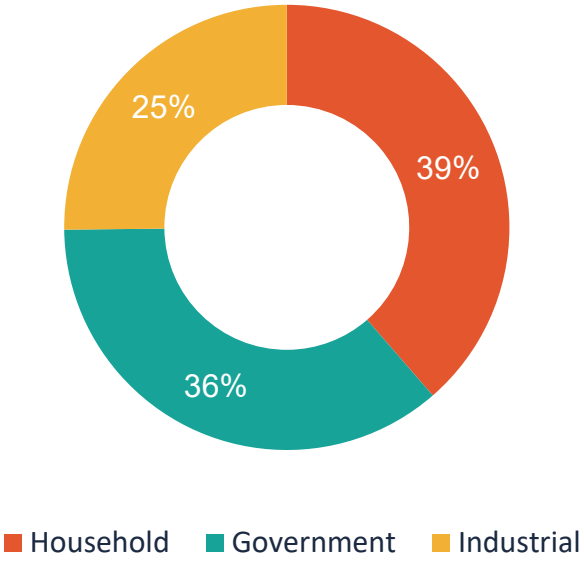
1.42M

Synthetic paint

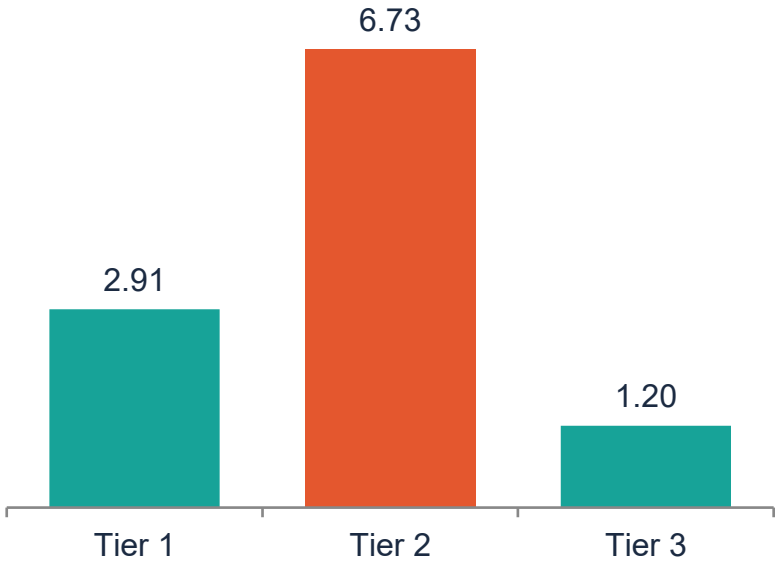
Lowest category sales

Where the sales come from

Sales by order type



Sales by city tier (millions)



Coupons drive purchases and first buys

Transactions: with vs without a campaign coupon

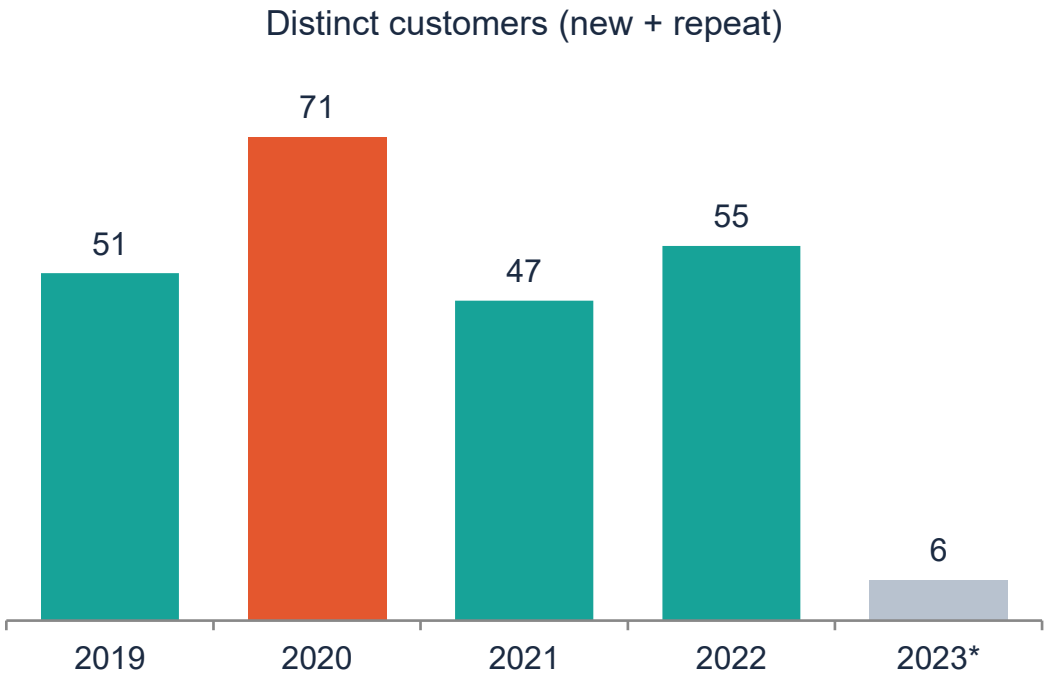


Customers: first purchase with vs without a coupon




```
First purchase flag: ROW_NUMBER() OVER (PARTITION BY Cust_Id ORDER BY PurchaseDate) = 1
```

Customer acquisition by year



88 Household customers

28 Industrial customers

17 Government customers

*2023 has March data only.

Market basket analysis

```
SELECT C1.item_id AS id1,
       C2.item_id AS id2,
       COUNT(C1.Trans_Id) AS freq
FROM sqlproject_customertransactiondata C1
JOIN sqlproject_customertransactiondata C2
  ON C1.Cust_Id = C2.Cust_Id
WHERE C1.Trans_Id != C2.Trans_Id
  AND C1.OrderType = C2.OrderType
  AND C1.item_id  != C2.item_id
GROUP BY C1.item_id, C2.item_id
HAVING COUNT(C1.Trans_Id) >= 3
ORDER BY freq DESC;
```

1. Self-join on the same customer
2. Same order type, different items
3. Keep pairs seen at least 3 times

Use: bundle offers and cross-sell for frequently co-bought paints.

Insights and recommendations

Sales are slipping

INSIGHT

Sales fell 11% in 2021 and 31% in 2022 after the 2020 peak.

ACTION

Launch a re-engagement coupon campaign for existing customers.

Q4 is the peak

INSIGHT

Q4 holds 43% of all transactions (130 of 300).

ACTION

Time campaigns ahead of Q4 to ride the seasonal demand.

Coupons and Tier 2

INSIGHT

Tier 2 gives 62% of sales; 67% of customers first bought with a coupon.

ACTION

Target acquisition coupons at Tier 2 and 3; bundle co-bought items.